Real Analysis: Exercise 3

Due date:

Question 1. What can you say about x and a if you are told that, for every $\epsilon > 0$, $|x - a| < \epsilon$? What if you are told that, for every positive integer n, $|x - a| < 1/n$?

Question 2. Suppose that $b < a < c$ and that f is a real-valued function defined for all $x \in (b, c)$. Suppose also that ℓ is a real number. What can you say about f if the following hold? (It might help to recall that a function is said to be *bounded* on the interval $[c, d]$ if there is a number $K > 0$ such that $-K \leq f(x) \leq K$ for all $c \leq x \leq d$.)

1. for all $\epsilon > 0$, there exists $\delta > 0$ such that $|f(x) - \ell| < \epsilon$ whenever $0 < |x - a| < \delta$.
2. for all $\delta > 0$, there exists $\epsilon > 0$ such that $|f(x) - \ell| < \epsilon$ whenever $0 < |x - a| < \delta$.
3. there exists $\epsilon > 0$, such that for all $\delta > 0$, $|f(x) - \ell| < \epsilon$ whenever $0 < |x - a| < \delta$.
4. there exists $\delta > 0$ such that for all $\epsilon > 0$, $|f(x) - \ell| < \epsilon$ whenever $0 < |x - a| < \delta$.

Question 3. Using the ϵ-δ definition of the limit, prove that $\lim_{x \to a} x = a$ and that $\lim_{x \to a} c = c$, where c is a constant.

Question 4. Using the ϵ-δ definition of the limit, prove that

1. $\lim_{x \to 2}(3x - 1) = 5$;
2. $\lim_{x \to 0} x^2 = 0$.

Question 5. Using the ϵ-δ definition of the limit, prove that $\lim_{x \to 3}(2x + 1) = 7$.

Question 6. Determine the following limits and prove that your answer is correct by directly appealing to the definition of the limit.

1. $\lim_{x \to \infty} \dfrac{3x^3 - 5x^2 + 7x - 13}{2x^3 - \pi}$.

2. $\lim_{x \to 1} \big(2x^2 - 3x + 5\big)$.

3. $\lim_{x \to 2} \dfrac{1}{1 - x}$.

4. $\lim_{x \to \infty} \dfrac{1}{x} \sin x$.

5. $\lim_{x \to 0} x \sin \dfrac{1}{x}$.